

OOAIA – Lab Assignment 02

Bookshelf Management using Linked Lists and Inheritance

Class Specification

Rajesh owns many books and wants to organize them using two different wooden bookshelves.

Bookshelf Types

1. VerticalBookshelf

- Open from one end only
- Books are added and removed only from the same end

2. HorizontalBookshelf

- Open from both ends
- Books are added from one end and removed from the opposite end

You must implement these bookshelves using **Linked Lists only** and apply **Inheritance** and **Function Overriding** concepts of Object-Oriented Programming.

Book Class

Create a class Book with the following private data members:

- `string title`
- `string author`
- `int year`
- `int price`

Base Class: Bookshelf

Create an **abstract base class** named Bookshelf.

Data Members

- `Node* head` – start of the linked list
- `Node* tail` – end of the linked list

Member Function

- `bool checkEmpty()` – returns true if the shelf is empty, otherwise returns false

Derived Classes

Create the following classes inherited from `Bookshelf`:

- `VerticalBookshelf`
- `HorizontalBookshelf`

All remaining functions must be implemented only in the derived classes using function overriding.

Member Functions

1. `void addBook(Book b)`

- Adds a book to the shelf

2. `void removeBook(int n)`

- Removes `n` books from the shelf
- Print removed book details
- Print `e` if the shelf is empty

3. `void list()`

- Prints all books currently present in the shelf in the order in which they would be removed
- Print one book per line
- Print `e` if empty

4. `void reverse()`

- Reverses the order of books present in the shelf
- Must be implemented using linked list manipulation only
- Print `e` if the shelf is empty

5. `void reverseK(int k)`

- Reverses the books in the shelf in groups of size `k`
- If the number of remaining books is less than `k`, those books must remain in their original order
- Must be implemented using linked list manipulation only
- Creation or deletion of nodes during this operation is not allowed
- Print `e` if the shelf is empty

Output Format

Whenever a book is printed, it must be printed in the following format:

title author year price

- Each book must be printed on a new line
- If an operation is performed on an empty shelf, print `e` on a new line
- `REVERSE` and `REVERSEK` do not print anything unless the shelf is empty

Constraints

- Use Linked Lists only
- Arrays are not allowed
- STL containers are not allowed
- Must use Inheritance and Function Overriding

Input Commands

`ADD x title author year price`

`REMOVE n x`

`LIST x`

`REVERSE x`

`REVERSEK k x`

where $x \in \{V, H\}$ and k is a positive integer.

Sample Test Case – Basic Operations

Sample Input

`ADD V B1 A1 2010 100`

`ADD V B2 A2 2011 200`

`ADD H C1 A3 2012 300`

`LIST V`

`REMOVE 1 V`

`LIST V`

`LIST H`

Sample Output

`B2 A2 2011 200`

`B1 A1 2010 100`

`B2 A2 2011 200`

`B1 A1 2010 100`

`C1 A3 2012 300`

Sample Test Case – reverseK Operation

Sample Input

```
ADD V B1 A1 2010 100
ADD V B2 A2 2011 200
ADD V B3 A3 2012 300
ADD V B4 A4 2013 400
LIST V
REVERSEK 2 V
LIST V
```

Sample Output

```
B4 A4 2013 400
B3 A3 2012 300
B2 A2 2011 200
B1 A1 2010 100
B3 A3 2012 300
B4 A4 2013 400
B1 A1 2010 100
B2 A2 2011 200
```